

STATEMENT OF PURPOSE

Applicant Name: Ayush Raj

CGPA: 8.93

Target Program: Amazon ML Summer School 2026

Date: June 14, 2026

My focus in Artificial Intelligence and Machine Learning lies in building production-grade applications that solve practical data problems. To gain hands-on experience, I developed **Smart Expense Tracker**, a full-stack Progressive Web App (PWA) engineered with **React (Vite)**, **FastAPI (ASGI)**, and **PostgreSQL (Supabase)** to automate personal financial tracking and analyze transaction datasets.

During this project, I implemented three distinct ML pipelines. First, I integrated Large Language Models (LLMs) via **Gemini and Groq APIs** to build a conversational assistant. I designed custom **system prompts** and **context constraints** to parse unstructured natural language chat inputs into **structured JSON schemas** matching our SQLAlchemy **ORM models**. Second, I set up a **Computer Vision (OCR) pipeline** to segment and extract transaction details from uploaded receipt photos. Third, I implemented a statistical **Z-Score anomaly detection algorithm** to flag outlier transactions. If a new transaction exceeded a threshold of **2.0 standard deviations** from the user's running historical average, the app triggered a budget alert. While integrating these APIs was a great exercise, it highlighted the limitations of relying on black-box third-party models, particularly regarding inference latency, data privacy, and the lack of custom fine-tuning capability.

This project helped me identify key gaps in my current machine learning knowledge. I want to move beyond calling third-party API endpoints and learn to train, evaluate, and deploy **custom models** myself. In computer vision, instead of generic OCR, I want to study **Convolutional Neural Networks (CNNs)** and **Vision Transformers (ViTs)** to build custom segmenting models that handle noisy, low-resolution receipt images. On the data side, simple statistical Z-scores assume a Gaussian distribution, which is rarely true for skewed transactional data. I want to master unsupervised anomaly detection algorithms, such as **Isolation Forests** and **Autoencoders**, to identify complex fraudulent patterns. Finally, I need to learn about **feature engineering**, handling imbalanced datasets, and deploying optimized models on cloud platforms like **AWS SageMaker**.

The Amazon ML Summer School (MLSS) 2026 is the perfect opportunity to bridge these gaps. Learning directly from Amazon's ML scientists about **deep learning theory, optimization, and NLP** matches my career goal of transitioning from a full-stack software developer to a specialized machine learning engineer. With a strong academic record (**CGPA: 8.93**) and practical deployment experience, I am ready for this academic challenge. I am confident this program will provide the mathematical foundations and engineering principles needed to build next-generation intelligent systems.